Panithan Lertsuntivit

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Education

University of California, Berkeley December 2025
B.S. Mechanical Engineering *GPA: 3.73*

Coursework: Mechatronic Design, Introduction to Flight Mechanics, Fluid Mechanics, Dynamic Systems and Feedback, Embedded Microprocessor Based Systems, Dynamics, Thermodynamics, Introduction to Solid Mechanics

Programs / Languages: SolidWorks, AutoCAD, Onshape, ESP-IDF, LabVIEW, Python, MATLAB, C

Professional Experience

UC Berkeley Mechanical Engineering August 2024 - Present
Teaching Assistant *Berkeley, CA*

- Collaborate regularly with the professor and other teaching assistants to ensure scheduled course progression
- Mentor over 300 students across ME 103 [Experimentation and Measurements] and ME 100 [Electronics for IOT] by hosting weekly discussions, clarifying topics in office hours, and designing enhancements for student assignments
- Guide weekly lab sections with hands-on experience using microcontrollers, electronic sensors and actuators

National Tsing Hua University June 2025 - August 2025
Yang Lab for Biomechanics | *Research Intern* *Hsinchu, Taiwan*

- Rapidly learned the principles of biomechanics for immediate contributions to active research through prototyping
- Iteratively designed and validated a fluid mixing adapter to produce a stable mixture comprised of ink and mucus
- Modeled and coded a controllable fluid release system to build a framework for iterative ink pattern testing

Organizations

Aerospace SAE at Berkeley September 2024 - Present
Director of Avionics *Berkeley, CA*

- Engineered and validated a thrust test rig for propeller and motor tests in static and dynamic wind conditions
- Implemented Molex connectors and integration geometries for secure electrical and physical connections; facilitating a sub-2-minute fully functional wing panel assembly on the flight line for the 2025 SAE Aero Design Competition

Projects & Interests

eV/STOL August 2025 - Present

- Engineer an aircraft capable of vertical / short take off and landing for increased maneuverability and efficiency
- Integrate electrical components to support a maximum take off weight of 18 N with a cruising speed of 13 m/s

Real-Time Plotter January 2024 - May 2024

- Developed a wireless ESP32-based plotter using Espressif's IoT Development Framework (ESP-IDF), capable of drawing geometric shapes at a frequency of 65 Hz directed from a LabVIEW graphical user interface
- Optimized system efficiency through multitasking between real-time motor actuation and wireless communication

Mobile Desk September 2023 - December 2023

- Designed and created a portable desk that conformed to GD&T standards for fits between part for integration
- Manufactured aluminum components using manual lathes and mills in the UC Berkeley Etcheverry Machine Shop

Wall Climbing Vehicle January 2023 - May 2023

- Designed a remote control vehicle capable of maneuvering on a vertical surface using SolidWorks and Onshape
- Prototyped various wooden living hinges, achieving rapid production of flexible laser-cut wood geometries

Fields of Interests: Robotics, Mechatronics, Mechanical Design
Hobbies: Exploring Cultural Landmarks and Establishments and Asian Gourmet Cooking